

Higher-order Cheeger inequalities for graphons

Mugdha Mahesh Pokharanakar

IISER Bhopal

Higher-order Cheeger inequalities for graphs

Let G be a graph having n vertices with $n \geq 2$. Assume that G has no isolated vertex. Dodziuk (1984), Alon–Milman (1985), and Alon (1986) proved the following.

Theorem 1 (Discrete Cheeger–Buser inequality)

Let λ_2 denote the second smallest eigenvalue of the Laplacian of G , and h_G denote its Cheeger constant. Then, we have

$$\frac{\lambda_2}{2} \leq h_G \leq \sqrt{2\lambda_2}.$$

In 2014, Lee, Oveis Gharan and Trevisan established the following, resolving a conjecture of Miclo (2008).

Theorem 2 (Higher-order Cheeger inequalities for graphs [LGT14])

Given any positive integer k , the k -th smallest eigenvalue λ_k of the Laplacian of G , and its k -way expansion constant $h_G(k)$ satisfy

$$\frac{\lambda_k}{2} \leq h_G(k) \leq O(k^2)\sqrt{\lambda_k}.$$

Graphons

A **graphon** W is a Lebesgue measurable function $W: [0, 1]^2 \rightarrow [0, 1]$ satisfying $W(x, y) = W(y, x)$ for all $x, y \in [0, 1]$.

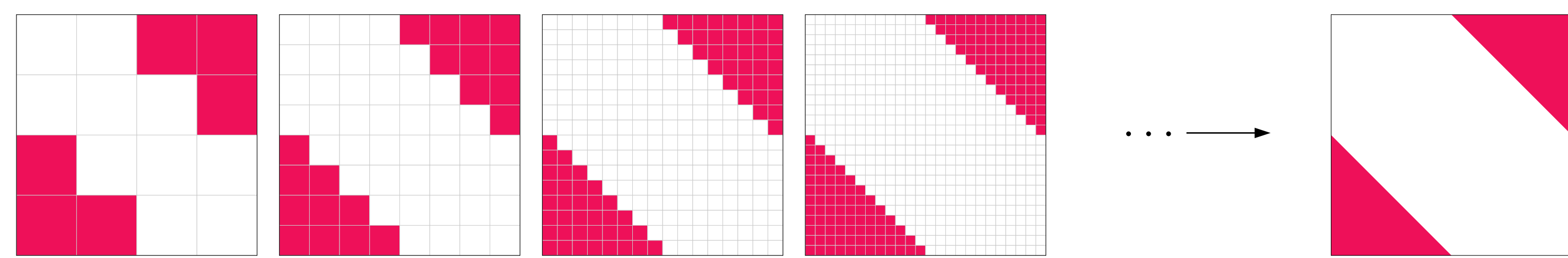
The theory of **graph limits** was developed by Borgs, Chayes, Lovász, Sós and Vesztergombi around 2006–2008. They proved that every graph limit can be represented by a graphon.

Let G be a graph on n vertices. Its **associated graphon** W_G is defined as follows. Consider the partition $\{P_1, \dots, P_n\}$ of the interval $[0, 1]$, where

$$P_k = \left[\frac{k-1}{n}, \frac{k}{n} \right) \text{ for every } 1 \leq k < n, \text{ and } P_n = \left[\frac{n-1}{n}, 1 \right].$$

For any $1 \leq i, j \leq n$ and $(x, y) \in P_i \times P_j$, define

$$W_G(x, y) = \begin{cases} 1 & \text{if there is an edge between } v_i \text{ and } v_j, \\ 0 & \text{otherwise.} \end{cases}$$



Cheeger–Buser inequality for graphons

Let μ_L denote the Lebesgue measure on $I := [0, 1]$. Let W be a connected graphon.

- For every measurable subset A of I , setting

$$\nu(A) = \int_{A \times I} W(x, y) dx dy,$$

defines a measure ν on I . The inner product on $L^2(I, \nu)$ is denoted by $\langle \cdot, \cdot \rangle_\nu$.

- Defining $\eta(S) = \int_S W(x, y) dx dy$ for every measurable subset S of I^2 , yields a measure η on I^2 .
- Let $E = \{(x, y) \in I^2 : y > x\}$. For any function $f: I \rightarrow \mathbb{R}$ and $(x, y) \in I^2$, let

$$df(x, y) = f(y) - f(x).$$

This defines a function $d: L^2(I, \nu) \rightarrow L^2(E, \eta)$, which is a bounded linear operator.

- The **Laplacian** $\Delta_W: L^2(I, \nu) \rightarrow L^2(I, \nu)$ of W is a self-adjoint, positive semidefinite linear operator, defined as $\Delta_W = d^*d$.
- The **bottom of the spectrum** of Δ_W is defined by

$$\lambda_W = \inf_{\substack{f \in L^2(I, \nu) \setminus \{0\} \\ \langle f, 1 \rangle_\nu = 0}} R_{\Delta_W}(f),$$

where 1 denotes the function on I taking value 1 μ_L -a.e., and $R_{\Delta_W}(f) = \frac{\langle \Delta_W f, f \rangle_\nu}{\langle f, f \rangle_\nu}$.

- The **Cheeger constant** of W is defined by

$$h_W = \inf_{\substack{A \subseteq I \text{ measurable} \\ 0 < \mu_L(A) < 1}} \max \left\{ \frac{\eta(A \times A^c)}{\nu(A)}, \frac{\eta(A^c \times A)}{\nu(A^c)} \right\}.$$

Khetan and Mj established the Cheeger–Buser inequality for graphons in 2024.

Theorem 3 (Cheeger–Buser inequality for graphons [KM24])

The bottom of the spectrum of the Laplacian of W and its Cheeger constant satisfy

$$\frac{\lambda_W}{2} \leq h_W \leq \sqrt{8\lambda_W}.$$

Higher-order Cheeger inequalities for graphons

Fix a connected graphon W , and a positive integer k .

Let $\mathcal{S}_k$ denote the set of all k -dimensional subspaces of $L^2(I, \nu)$. Define

$$\lambda_W(k) = \inf_{S \in \mathcal{S}_k} \max_{f \in S \setminus \{0\}} R_{\Delta_W}(f).$$

Let $\mathcal{P}_k$ denote the set of all k -tuples $(A_1, \dots, A_k)$ of measurable subsets of I satisfying $\nu(A_i) > 0$ for all i , and $A_i \cap A_j = \emptyset$ whenever $i \neq j$.

The **k -way expansion constant** $h_W(k)$ of W is defined by

$$h_W(k) = \inf_{(A_1, \dots, A_k) \in \mathcal{P}_k} \max_{1 \leq i \leq k} \frac{\eta(A_i \times A_i^c)}{\nu(A_i)}.$$

Theorem 4 (Higher-order Cheeger inequalities for graphons [Pok25])

The k -way expansion constant $h_W(k)$, and $\lambda_W(k)$ satisfy

$$\frac{\lambda_W(k)}{2} \leq h_W(k) \leq O(k^{3.5})\sqrt{\lambda_W(k)}.$$

Our proof is inspired from ideas in [LGT14] and [Tre17, Chapters 7,8].

Sketch of the proof of the bound $h_W(k) \leq O(k^{3.5})\sqrt{\lambda_W(k)}$

Let $\{f_1, \dots, f_k\}$ be an orthonormal subset of $L^2(I, \nu)$. Define $F: I \rightarrow \mathbb{R}^k$ as

$$F(x) = (f_1(x), \dots, f_k(x)) \text{ for all } x \in I.$$

For any $x, y \in I$ with $F(x), F(y) \neq 0$, define $d_F(x, y) = \left\| \frac{F(x)}{\|F(x)\|} - \frac{F(y)}{\|F(y)\|} \right\|$.

For any measurable subset A of I , let $\text{mass}_F(A) = \int_A \|F(x)\|^2 d\nu(x)$.

We partition $\mathbb{R}^k$ into cubes of equal size, having a fixed **small** side length, and remove from each cube, a fixed **very small** amount of region near its boundary, to get **smaller**, and **well-separated** cubes of equal size. Considering **random** translates of these cubes under a **suitable** probability distribution, we prove that there is an element u of $\mathbb{R}^k$ such that intersecting $\mathbb{S}^{k-1}$ with the u -translates of these cubes yields the lemma below.

Lemma 5 (“Small” and “well-separated” sets with “large” total F -mass)

For some $m \geq 1$, there exist pairwise disjoint measurable subsets $T_1, \dots, T_m$ of I such that the following holds.

- $\sum_{i=1}^m \text{mass}_F(T_i) \geq k - \frac{1}{4}$,
- for any $1 \leq i, j \leq k$ with $i \neq j$, if $x \in T_i$ and $y \in T_j$, then $d_F(x, y) \geq \frac{1}{4\sqrt{5}k^3}$,
- $\text{mass}_F(T_i) \leq 1 + \frac{1}{4k}$ for all $1 \leq i \leq m$.

After **merging** some of the sets as above, we obtain the following.

Lemma 6 (k -many “well-separated” sets with “sufficient” F -mass)

There exist pairwise disjoint measurable subsets $A_1, \dots, A_k$ of I such that

- for any $1 \leq i \leq k$, $\text{mass}_F(A_i) \geq \frac{1}{2}$,
- for any $1 \leq i, j \leq k$ with $i \neq j$, if $x \in A_i$ and $y \in A_j$, then $d_F(x, y) \geq \frac{1}{4\sqrt{5}k^3}$.

Considering **smoothened indicator functions** of the sets as above, and scaling them by $\|F\|$, gives us the lemma below.

Proposition 7 (“Localization” of F on k -many “well-separated” sets)

There exist disjointly supported functions $g_1, \dots, g_k \in L^2(I, \nu) \setminus \{0\}$ satisfying

$$R_{\Delta_W}(g_i) \leq O(k^7) \max_{1 \leq j \leq k} R_{\Delta_W}(f_j)$$

for all $1 \leq i \leq k$.

Combining Proposition 7 with Lemma 8 proves the desired upper bound on $h_W(k)$.

Lemma 8 (Existence of a set with “small” expansion)

For any element g of $L^2(I, \nu) \setminus \{0\}$, there exists a measurable subset A of I with $A \subseteq \text{supp}(g)$ and $\nu(A) > 0$, satisfying the inequality

$$\frac{\eta(A \times A^c)}{\nu(A)} \leq \sqrt{2R_{\Delta_W}(g)}.$$

References

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- [Pok25] Mugdha Mahesh Pokharanakar, *Higher-order Cheeger inequalities for graphons*, 2025. Available at <https://arxiv.org/abs/2511.07016>.
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